ReCycleGuard

Agentic AI for E-Waste EPR Compliance Reconciliation

Agentic AI (26CD7PEAAI) · B.E. CSE (Data Science), VII Semester, BMSCE
Instructor: Dr. Kalyan Nagaraj

01 AREA, PROBLEM AND SCOPE

Area: Circular economy and environmental compliance; India's e-waste Extended Producer Responsibility (EPR) reconciliation. Producer obligations and supporting records can disagree because of duplicates, quantity/category mismatches, incorrect periods, missing evidence or outdated rule references. Manual reconciliation must distinguish record errors from an unresolved shortfall.

Scope: One producer, one reporting period, limited equipment categories, structured records and a curated regulatory corpus. Use synthetic or de-identified data. Exclude live portal integration, regulatory filing and automatic legal compliance determination.

02 DECISION SUPPORTED AND OUTPUT

Which discrepancy should the compliance officer investigate next, and what verification or corrective action does the evidence support?

An auditable report will link findings to record IDs, calculations, applicable provisions/versions, open questions and next steps. Internal statuses: *Reconciled; Discrepancy detected; Pending verification; Requires expert review.* These are proposed system statuses, not official regulatory classifications.

03 AGENTIC CONCEPT MAPPING

Planning/reasoning	Plan checks → use tools → inspect results → investigate gaps → revise or stop within configured step limits.
Tool use	Code checks duplicates, categories, quantities and periods; calculates the apparent gap from configured obligations and verified eligible credits.
Agentic RAG	Select period-relevant provisions, assess evidence, re-query weak results and escalate unresolved applicability; retain source/version references.
Observability	Trace actions, tool inputs/outputs, retrieved evidence, adaptations, stop reasons and final findings for every run.
Evaluation	Golden cases; measure discrepancy precision/recall, arithmetic accuracy, citation support and appropriate escalation.
Bonus concepts	Collector/auditor collaboration, human review, input/output validation and bounded loops.

Agentic fit: The agent chooses follow-up checks and retrieval in response to missing or conflicting evidence, rather than stopping at a fixed calculation or single prompt.

04 DATA, TOOLS AND FEASIBILITY

Data: Obligation tables; certificate/transaction CSVs with IDs, quantities/units, categories, periods and allocation links; official E-Waste (Management) Rules, 2022, applicable amendments and CPCB guidance, tagged by version and effective period. Confirm document applicability during design.

Proposed tools: Python/pandas, SQLite, LangGraph, a searchable document index, a bounded LLM interface and structured traces. Regulatory sources: CPCB, EPR portal and MoEFCC. **Experiment:** Compare fixed retrieval with agent-controlled re-querying on identical cases; measure evidence quality, detection and tool/call cost. Results remain to be measured.

05 TEAM AND PROPOSED OWNERSHIP

Member	Student ID	Responsibility
R V Abhishek **(Lead)**	1BM23CD047	Coordination/submissions; orchestration and integration.
H Sampreeth Bhat	1BM23CD020	Regulatory corpus, retrieval, temporal metadata and citations.
Sushanth S	1BM23CD063	Data contracts, reconciliation tools, golden cases and evaluation.
Vedantha Vinay Vernekar	1BM23CD069	Tracing, experiment, report/demo and human review flow.

Component responsibilities are proposed; every member must explain the whole system.

MILESTONES R2: 24 Oct · R3: 31 Oct · R4: 7 Nov · Final submission: 12 Nov 2026

REVIEW BOUNDARY A numerical gap triggers investigation. Uncertain evidence requires abstention or expert review. Outputs remain advisory and human-confirmed.